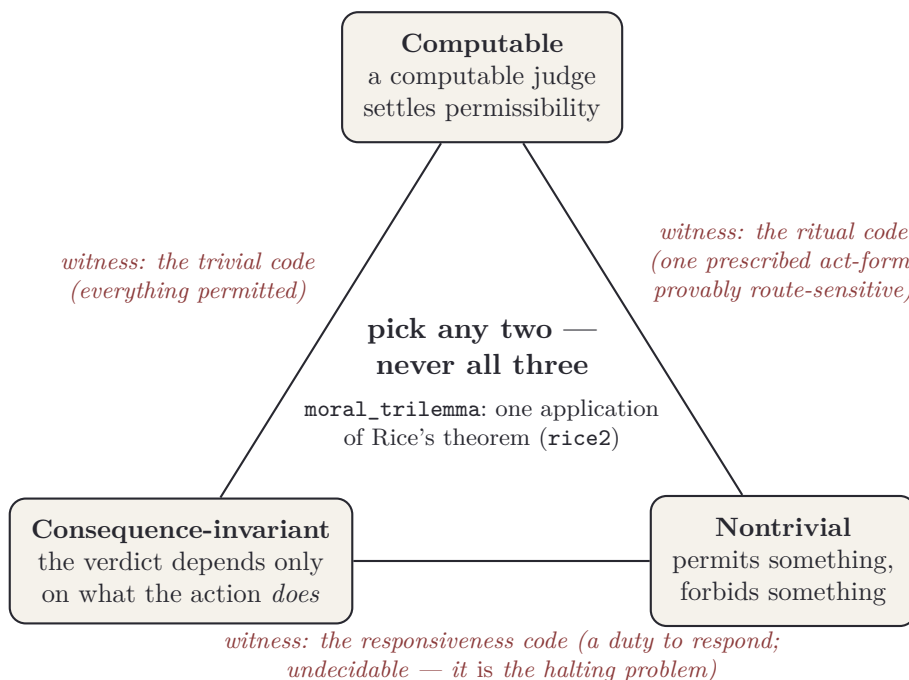


The Moral Trilemma

Rice's theorem, ported to ethics

machine-checked in Mettapedia/Ethics/MoralUndecidability.lean



Setting. Actions are programs (partial recursive codes); an action's *consequence profile* is the partial function it computes over situations; a moral code is a set of permissible actions. **Theorem.** No moral code is simultaneously computably judgeable, consequence-invariant, and nontrivial. Every pair is jointly satisfiable, witnessed as above. **Corollary** (`decidable_nontrivial_route_sensitive`). Decidability *forces* route-sensitivity: a workable moral code must distinguish some actions with identical consequences — judging the act rather than the outcome is not deontological taste but the only computable option. **Moral halting theorem** (`responsivenessCode_undecidable`). The duty to determine whether an action will respond cannot be computably audited.

“Rice’s theorem walks into moral philosophy wearing no disguise at all. Any computable judge whose verdict depends only on what actions do must permit everything or nothing. And the halting problem arrives as the duty to respond: the moral law that you must answer is precisely the law that no machine can check. The best jokes are the ones the universe writes itself.”

— imagined Raymond Smullyan, delighted

Verified in Lean 4 over Mathlib’s computability library; no additional axioms. Companion results: consequentializing, deontizing, and virtualizing are each *universal* at the price of a gerrymandered presentation, and undecidability transfers along every faithful computable translation between paradigms. Complexity-theoretic context: J. Stenseke, *On the computational complexity of ethics*, AI Review (2024).